SY B.Tech. – ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

PRACTICAL ASSIGNMENT TEMPLATE

1. Student Details

Student Name	Pranav Chopade
PRN	202501110025
Division	A
Batch	A1
Roll No.	202501110025
Date of Submission	23-10-2026

2. Practical Details

Practical No.	5
Practical Title	CLASSES, OBJECTS, THIS POINTER, INLINE FUNCTION, STATIC MEMBER FUNCTION AND FRIEND CLASS
Date Performed	23-10-2026
Faculty Name	Dr. Khushal Khairnar

3. Problem Statement

Write a C++ program demonstrating classes and objects, the this pointer, an inline member function, a static member function, and a friend class. Use a Student class to store student details, count created objects, display details, and allow a friend class to access private data.

4. Objectives

1. Understand classes and objects in C++.
2. Use the this pointer to refer to the current object.
3. Implement inline and static member functions.
4. Demonstrate private data access through a friend class.

5. Expected Learning Outcomes

After completing this practical, the student will be able to:

1. Create classes and objects with data members and member functions.
2. Use this pointer to refer to the current object.
3. Define and call inline and static member functions.
4. Explain how a friend class can access private members.

6. Theory / Concept

Class: A user-defined data type that groups data members and member functions.

Object: An instance of a class.

this Pointer: A pointer in non-static member functions that refers to the current object.

Inline Function: A function declared inline; the compiler may substitute its body at the call site.

Static Member Function: A class function that can be called using the class name and directly accesses static members.

Friend Class: A class declared as a friend can access private and protected members of the class that grants friendship.

7. Algorithm / Pseudocode

START

1. Define Student with private roll number and name members.
2. Declare a static counter to track Student objects.
3. Initialize object data using the constructor and this pointer.
4. Define an inline display() function and static showCount() function.
5. Declare StudentDetails as a friend class and access Student private data through it.
6. Create two Student objects in main().
7. Display both students, the object count, and private data through the friend class.

STOP

8. Technology / Tools / Libraries Used

Programming Language	C++
IDE / Platform	Dev C++
Libraries / Packages	iostream, string

9. Implementation / GitHub Repository

GitHub Repository URL	https://github.com/pranav-4797/OOP-Practical-Assignment
Repository Name	OOP-Practical-Assignment
Practical Folder	Practical 5
Branch Used	main
Commit ID / Latest Commit	Not provided

Repository Contents Used

The source file is provided separately as “Practical_5_Class_Objects.cpp”. Add it to the Practical 5 folder in the repository if required. The output image illustrates the expected sample output.

10. Test Cases / Input and Output

Test Case	Action	Expected Output	Actual Output	Status
1	Create s1(101, Aarav)	Roll 101; Name Aarav	Matches sample	Pass (sample)
2	Create two objects	Object count = 2	2	Pass (sample)
3	Friend class displays s1	Private details displayed	Matches sample	Pass (sample)

11. Results / Observations

The program demonstrates classes and objects, initializes members using this, displays details through an inline function, counts objects using a static member and function, and accesses private student data through a friend class.

12. Source Code

```
#include <iostream>
#include <string>
using namespace std;

class Student {
private:
int rollNo;
string name;
static int count;

public:
Student(int roll, string studentName) {
this->rollNo = roll;
this->name = studentName;
count++;
}

inline void display() {
```

```
cout << "Roll No: " << rollNo << endl;
cout << "Name: " << name << endl;
}

static void showCount() {
cout << "Total student objects: " << count << endl;
}

friend class StudentDetails;
};

int Student::count = 0;

class StudentDetails {
public:
void show(Student s) {
cout << "Accessing private data through friend class:" << endl;
cout << "Roll No: " << s.rollNo << endl;
cout << "Name: " << s.name << endl;
}
};

int main() {
Student s1(101, "Aarav");
Student s2(102, "Priya");

cout << "Student 1:" << endl;
s1.display();

cout << "\nStudent 2:" << endl;
s2.display();

cout << endl;
Student::showCount();

cout << endl;
StudentDetails details;
details.show(s1);

return 0;
}
```

Screenshots / Output

Practical 5 - Classes and Objects - Output

```
Student 1:
Roll No: 101
Name: Aarav

Student 2:
Roll No: 102
Name: Priya

Total student objects: 2

Accessing private data through friend class:
Roll No: 101
Name: Aarav
```

Sample run: two Student objects are created, displayed, counted, and accessed by the friend class.

13. Conclusion

This practical demonstrates classes and objects, the this pointer, an inline member function, a static member function, and a friend class. The output shows both student records, the object count of two, and private data accessed through the friend class.

14. Assignment Questions

Q1. What is a class?
Answer: A user-defined type that groups data and functions.

Q2. What is the use of this pointer?
Answer: It refers to the current object.

Q3. What is a static member function?
Answer: A function belonging to the class that can be called with the class name and directly accesses static members.

Q4. What is an inline function?
Answer: A function declared with inline; the compiler may substitute its body at the call site.

Q5. What is a friend class?
Answer: A class allowed to access private and protected members of the class that declares it as a friend.

15. Student Declaration

I hereby declare that the above practical work has been completed by me and that the implementation, analysis, and answers are my own work.

Student Name	Pranav Chopade
PRN	202501110025
Signature	